Cross-City Urban Heat Hotspot Prediction

Two Ways to Evaluate Hotspot Prediction

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STAT 5630 Final Project | April 2026

Research Question + Validation Design

Predictor Set

- impervious
- land cover
- elevation
- water dist.
- NDVI
- climate

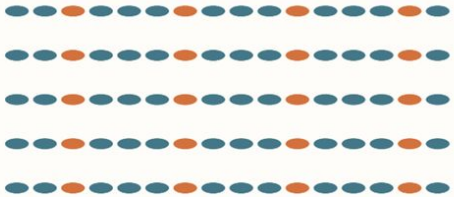
→

Hotspot Risk

top-decile model score

Within-City Held-Out Cells

Train and test cells come from every city.

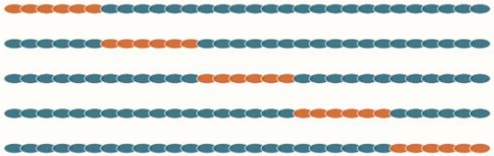


Train

Held Out

City-Held-Out Transfer

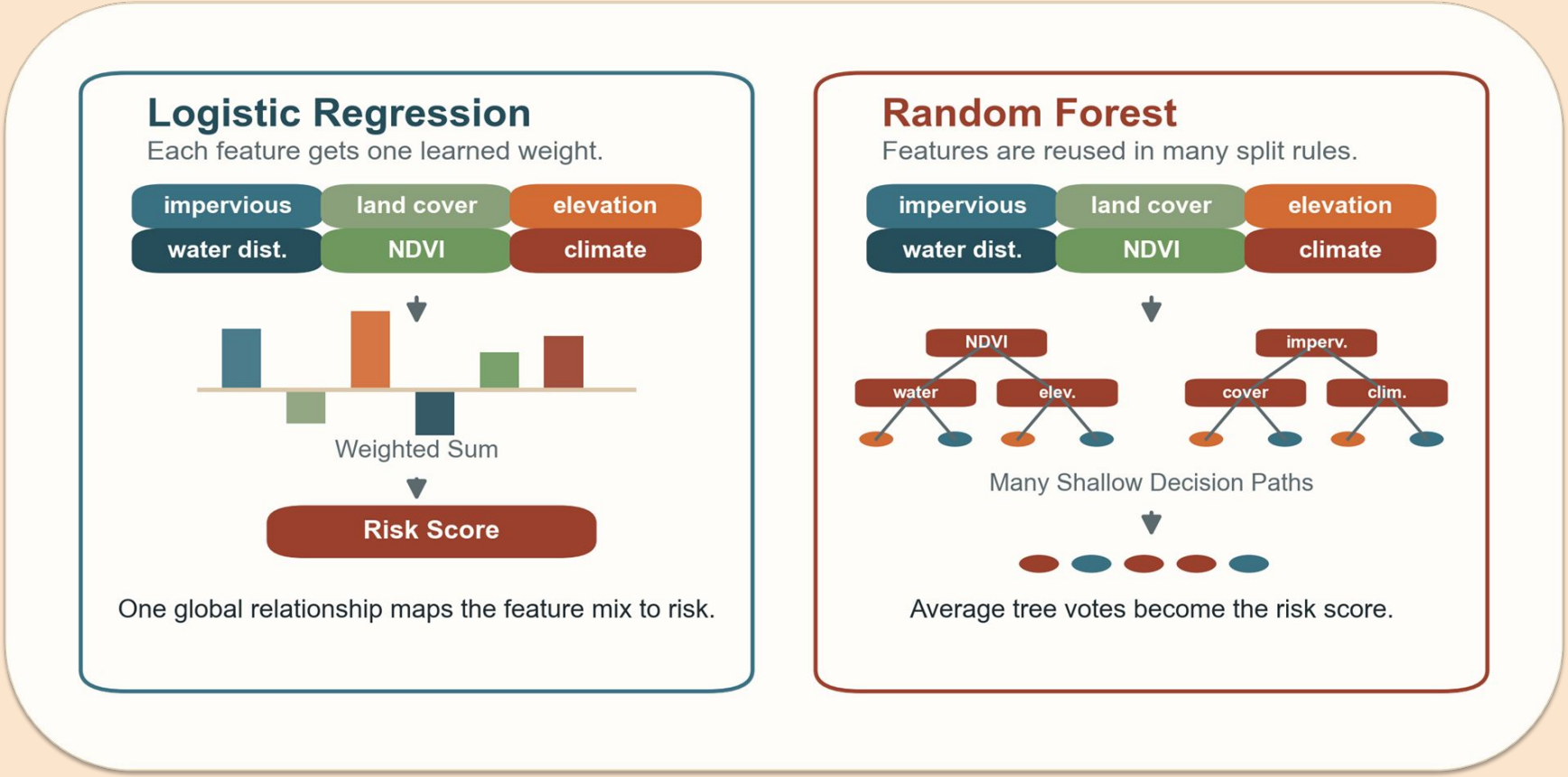
5 outer folds; 6 unseen cities per fold.



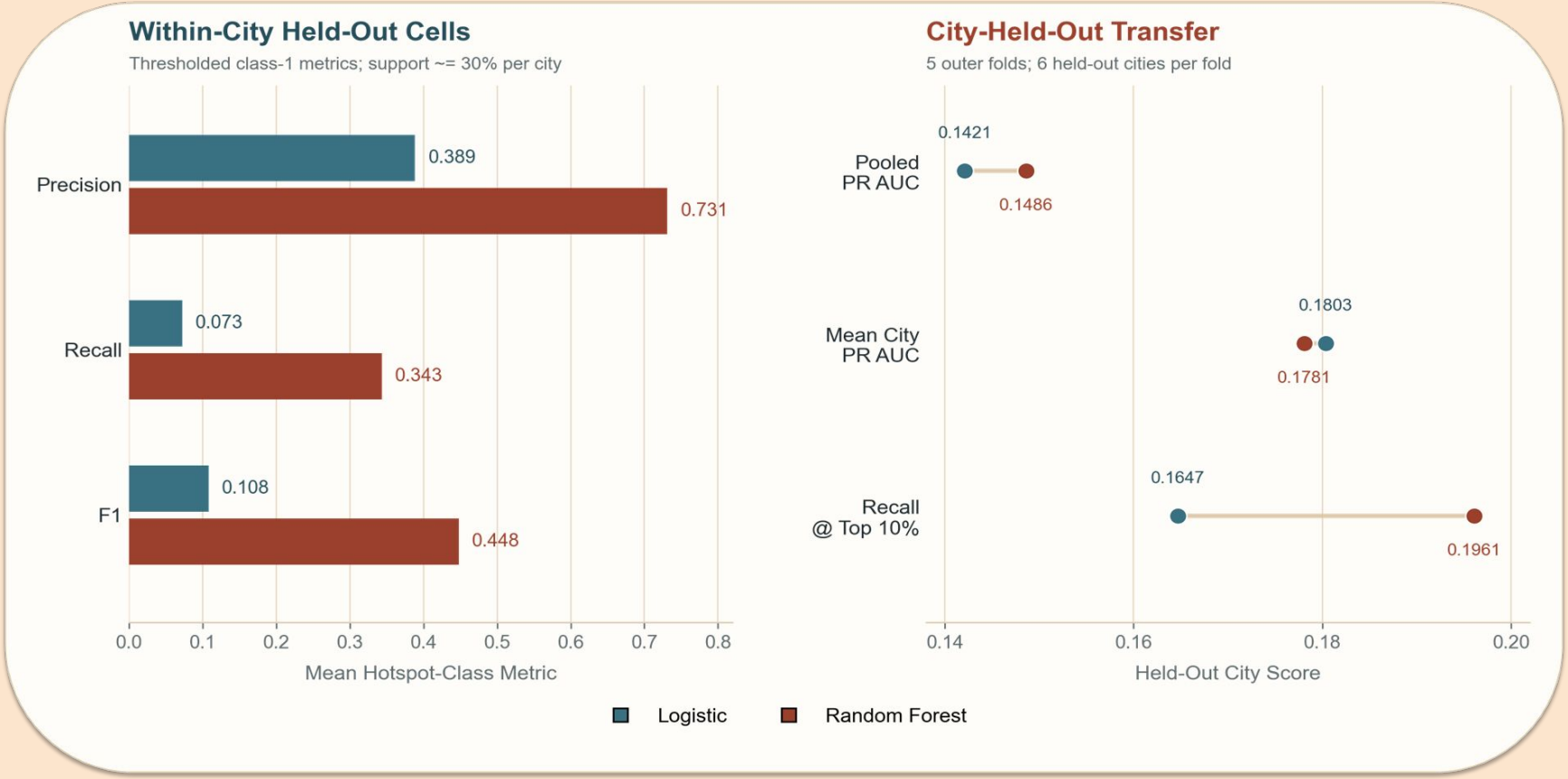
Seen

Unseen

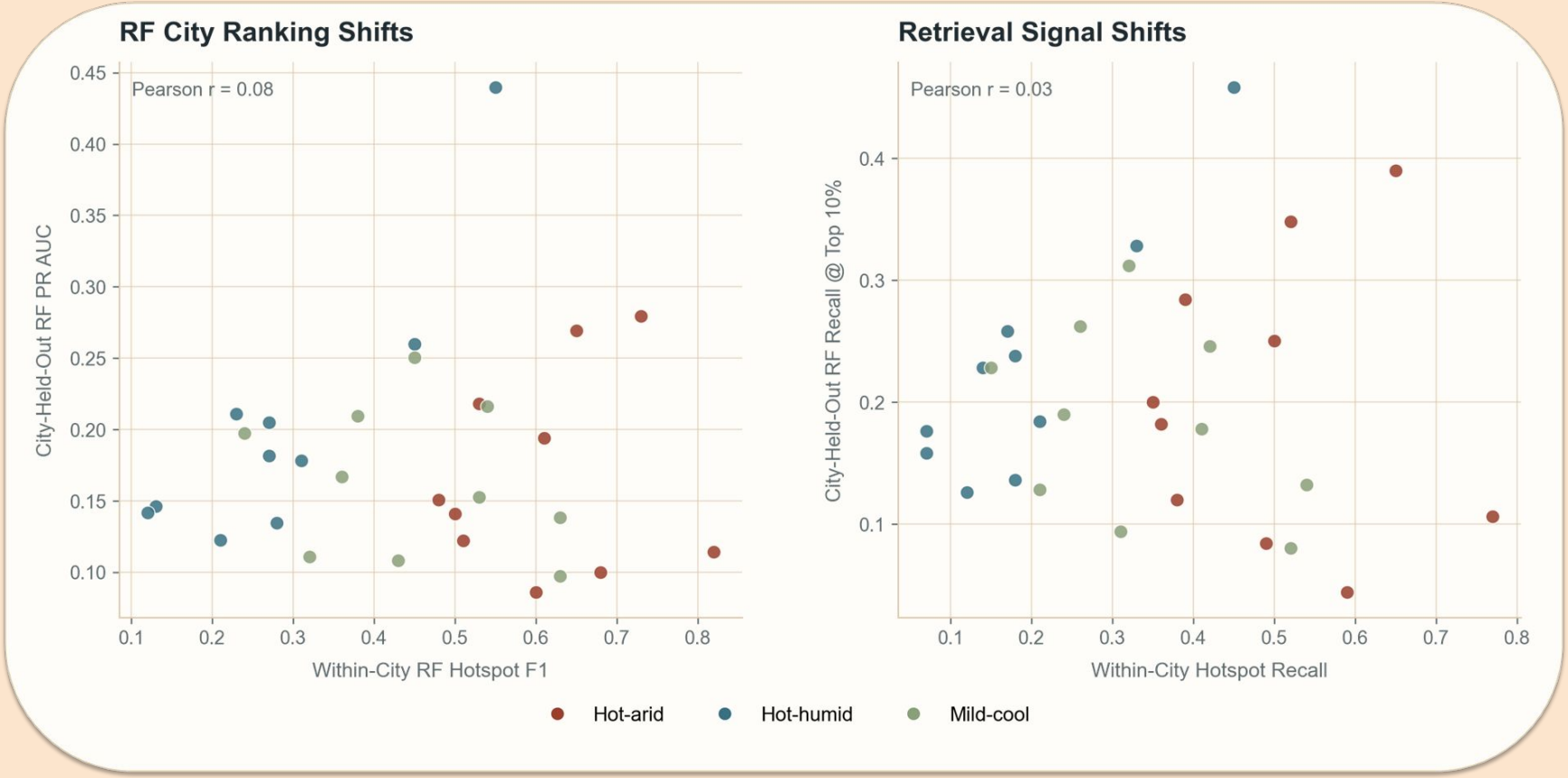
Modeling Selection: Logistic vs Random Forest



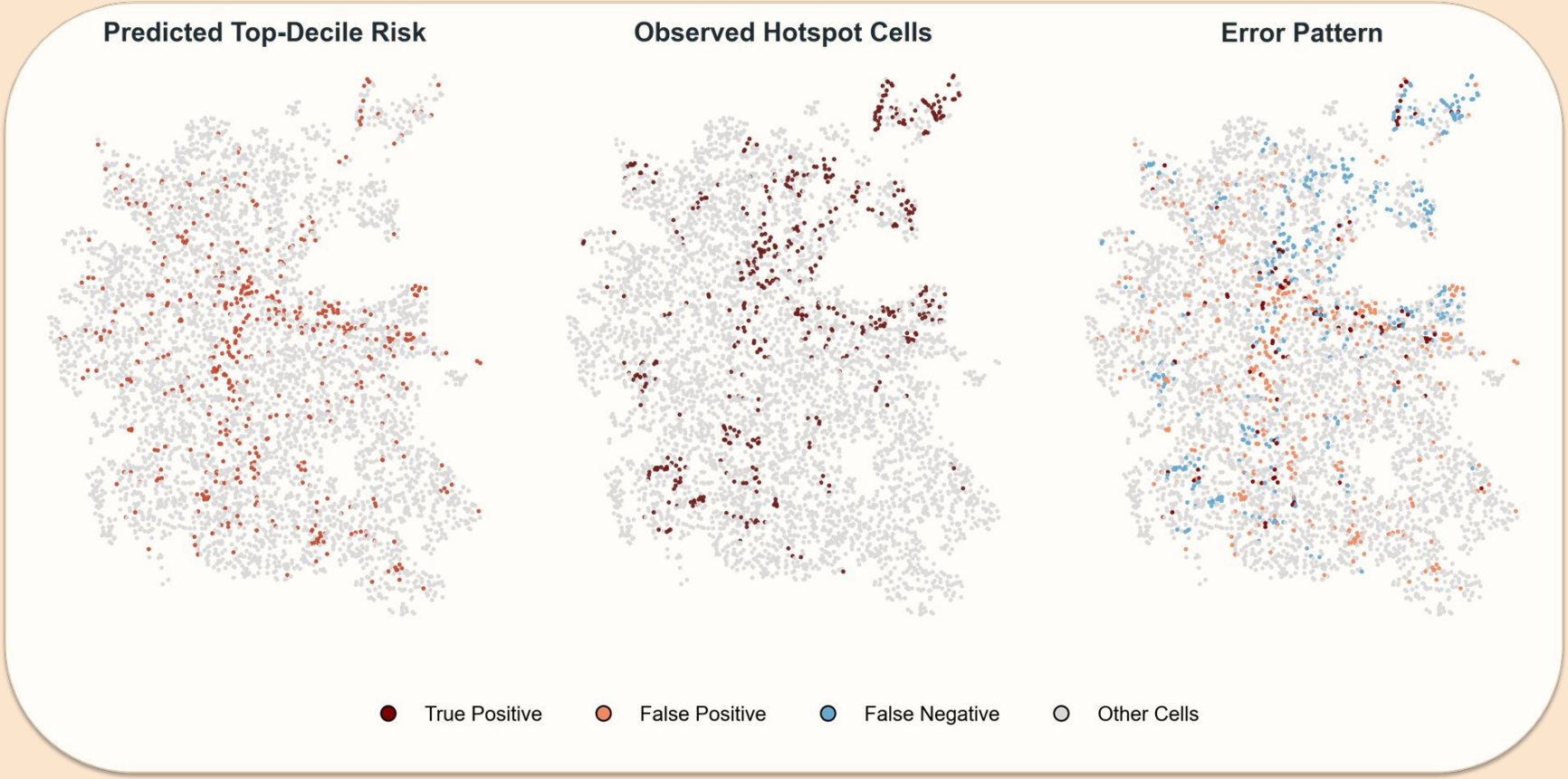
Results Side by Side



City-Level Signal Shifts Across Evaluation Designs



Held-Out Denver Map Example



Any Questions?